BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 096 · Primary Storage: RAM, ROM and Cache

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. After giving a technical point in an explain answer, what should follow?

Q2. A = 1 and B = 0. What is A AND B?
A. 1
B. 2
C. 10
D. 0

Q3. A truth table has independent inputs A, B and C. Give the number of input combinations and list them in ascending binary order.

Q4. A loop should run while finished is false. It uses finished as its condition. Correct the condition. Explain what happens when finished=false and finished=true.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

RAM — Volatile primary storage used for active programs and data.

ROM — Non-volatile memory typically storing firmware.

Cache — Small, fast memory close to the CPU.

Volatile — Loses stored contents when power is removed.

New learning

RAM is working memory: RAM stores programs and data currently in use. It is volatile: its contents are lost when power is removed.

ROM and startup: ROM is non-volatile and typically stores essential firmware or startup instructions.

Why cache exists: Cache is small, very fast memory close to the processor, holding frequently or recently used data and instructions.

Compare the roles: RAM is general working memory; ROM retains firmware; cache reduces repeated slower memory accesses.

Worked example

A laptop uses RAM, ROM and cache. Explain the role of each and why cache helps the CPU. [6 marks]

1. Identify the memory purpose

2. RAM stores active programs/data; ROM retains firmware across shutdown.

3. Cache holds frequently used instructions/data close to CPU, reducing RAM access.

4. RAM and cache are volatile; ROM retains contents without power. [2 marks for each correctly explained role]

We Do · Practise with guidance

Guided 1

Explain why opening more applications increases demand for RAM. [4]

Worked method: Each open program requires instructions/data in RAM [1]. More active applications occupy more RAM [1]. Available RAM decreases [1], potentially requiring virtual memory [1].

Guided 2

A device must retain startup code when unplugged. Explain why ROM is suitable. [4]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Worked method: ROM retains its contents without power [1], so firmware remains available after shutdown [1]. Startup instructions can be accessed when switched on [1], allowing the device to initialise [1].

Guided 3

Explain why CPU cache is usually smaller than RAM but helps performance. [4]

Worked method: Cache provides faster access than RAM [1] for frequently needed data [1], reducing time waiting for memory [1]. It is more expensive per unit of storage, so is generally smaller [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Change the comparison so the application only loads if it has enough free RAM.

```
freeMB = 800
neededMB = 800
if freeMB > neededMB then
    print("Load")
else
    print("Wait")
endif
```

iterate

Question 1

State whether RAM is volatile and describe what happens to its contents when power is lost. [2] Source: Original OCR J277-style practice question.

Question 2

Explain why a CPU benefits from cache memory. [3] Source: Original OCR J277-style practice question.

Question 3

Compare RAM and ROM using purpose and volatility. [4] Source: Original OCR J277-style practice question.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

An algorithm only gives reliable results when its inputs are handled as specified.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A linked reason or effect that answers the stated situation.

Q2. D — 0

Q3. 8: 000, 001, 010, 011, 100, 101, 110, 111.

Q4. Use NOT finished. It is true when finished is false, so the loop continues. It is false when finished is true, so the loop stops. The original does the reverse.

I Do / We Do pages · Guided practice

Q1. Each open program requires instructions/data in RAM [1]. More active applications occupy more RAM [1]. Available RAM decreases [1], potentially requiring virtual memory [1].

Q2. ROM retains its contents without power [1], so firmware remains available after shutdown [1]. Startup instructions can be accessed when switched on [1], allowing the device to initialise [1].

Q3. Cache provides faster access than RAM [1] for frequently needed data [1], reducing time waiting for memory [1]. It is more expensive per unit of storage, so is generally smaller [1].

You Do pages · Code First

```
freeMB = 800
neededMB = 800
if freeMB >= neededMB then
    print("Load")
else
    print("Wait")
endif
```

Expected result: Load

You Do pages · Independent questions

Q1. Volatile [1]; contents are lost when power is removed [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Q2. Frequently used data/instructions stored in cache [1]; CPU can access them faster than RAM [1], improving speed/reducing waits [1].

Q3. RAM stores currently running programs/data [1] and is volatile [1]; ROM typically stores firmware/startup instructions [1] and is non-volatile [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: